Aakanx Panchal

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Education

Honours Bachelor of Science (2017 – Present)
Integrated Science (iSci) Program (Biology)
McMaster University, Hamilton, ON

- Currently enrolled in McMaster University's 4-year Integrated Science program, an award-winning, limited enrollment program (60 students/year), which employs both traditional lecture-style and innovative project-based learning, while focusing on the development of self-directed learning skills in a supportive and collaborative environment
- ISCI 2A18 5-week Enrichment Project (Drug Discovery): Attacking chemotherapy resistance using DNA nanotechnology (Supervisor: Dr. Rosa da Silva)
- ISCI 3A12 13-week Independent Project: Characterizing symbiotic phenotypes of rhizobia harbouring minimal symbiotic genomes (Supervisors: Dr. Barney A. Geddes, Dr. Turlough M. Finan)
- ISCI 4A12 Honours Thesis: Expanding the legume host range of the nitrogen-fixing bacterium *Ensifer meliloti* (Supervisor: Dr. Turlough M. Finan)

International Baccalaureate (IB) Diploma Program (2013 – 2017)
Turner Fenton Secondary School, Brampton, ON

- Completed requirements of both Ontario Secondary School Diploma and International Baccalaureate Diploma
- IB Diploma Program courses: Higher Level - Chemistry, History, and English Literature; Standard Level - Biology, Mathematics, and French; Theory of Knowledge (ToK) course about philosophy and critical-thinking
- Investigated "Stalin and the Outbreak of the Cold War: 1945-1949" as part of the Extended Essay research paper component

Awards

- 1st Place Award,
  SciNapse Undergraduate Science Case Competition,
  University of Ottawa, 2018
- Deans' Honour List,
  McMaster University, 2017-2018, 2018-2019, 2019-2020
  - Based on a minimum 9.5/12.0 GPA on at least 27/30 units of courses
- $1000, McMaster Honour Award,
  McMaster University, 2017
  - For final admission average of 90-94.99%
- Musician of the Year,
  Turner Fenton Secondary School, 2016-2017
- Academic Excellence Award,
  Turner Fenton Secondary School, 2015-2016
- Jazz Ensemble Honour Award,
  Humber College Community Music School, 2016
- Exceptional Leadership Award,
  Peel Music Teachers' Association Music Festival, 2015

Publications

Journal Articles

1. (submitted) Geddes, B. A., Kearsley, J., Huang, J., Zamani, M., Muhammed, Z., Sather, L., Panchal, A., diCenzo, G. C., and Finan, T. M. Minimal gene set from *Sinorhizobium (Ensifer) meliloti* pSymA required for efficient symbiosis with *Medicago*. *Proceedings of the National Academy of Sciences of the United States of America.* Revision submitted 2020-11-11 (Manuscript # 2020-18015R).

Undergraduate Conference Abstract

1. An, Z., Grala, K., Panchal, A., and Trivedi, K., 2018. Investigating graphene oxide permeable reactive barriers for filtering groundwater contaminated from hydraulic fracturing. *University of Ottawa Science Undergraduate Research Journal*, [online] 1(1), pp.55–55. Available at: <https://uottawa.scholarsportal.info/ottawa/index.php/OSURJ-JRSUO/article/view/3705>.

Undergraduate Book Sections

2. Panchal, A., Romano, M., Singh, J., Somani, A., and Winter, M., 2019. Fighting Cancer With Wine: Nanoparticle-Conjugated Resveratrol Particles in Wine as Potential Chemotherapeutic Agents. In: *Terroir 2019*. [online] pp.37–42. Available at: <http://hdl.handle.net/11375/25404>.

1. (in print) Panchal, A. and Somani, A. Scientific Hypotheses in Collision: Dinosaur Extinction (K-Pg Extinction by Asteroid Impact). In: *History of the Earth Vol 9*. [online] pp.117–122.

iSci Blog: Synopsis

9. Panchal, A., 2020. Soil bacteria to the rescue. *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=64166>
8. Panchal, A., 2019c. The universal language. *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=62521>.
7. Panchal, A., 2019b. Plants listen to music too. *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=62524>.
6. Panchal, A., 2019a. Dinosaurs Ate Rocks! *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=62526>.
5. Panchal, A., 2018d. The Ethical and Biological Journey of the Human Genome Project. *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=62536>.
4. Panchal, A., 2018c. Santa's Companions Have UV Vision. *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=62528>.
3. Panchal, A., 2018b. Insights On Cell Survival Mechanisms Close To Home. *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=62532>.
2. Panchal, A., 2018a. Glia: the "other" cells in neuroscience. *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=62530>.
1. Panchal, A., 2017. Sleep–Health–Academics. *syn·op·sis*. Available at: <http://synopsis.mcmaster.ca/?p=62540>.

Research Experience

Research Student *(January 2020 – Present)*
McMaster University, Hamilton, ON

- Conducting research in Dr. Turlough M. Finan's lab group in the Department of Biology, studying symbiotic genes of the bacterium *Ensifer meliloti*
- Investigating symbiotic N_2-fixation phenotypes via shoot dry weight, nodule formation kinetics, competition for nodule occupancy, acetylene reduction assays
- Constructing bacterial strains using yeast recombineering along with more traditional molecular biology and microbiology techniques
- Exploring symbiotic effectiveness of rhizobia on numerous legume hosts such as *Medicago sativa* (alfalfa), *Vigna unguiculata* (cowpea), and *Phaseolus vulgaris* (common bean)

SciNapse Undergraduate Science Case Competition *(March 2018)*
University of Ottawa, Ottawa, ON

- 1st place winner at final round / national level of competition based on research proposal and poster presentation, competing with 122 other teams
- Collaborated in science research competition with three other team members, tackling task of improving filtration of contaminants from hydraulic fracturing, with guidance from Dr. Carolyn H. Eyles, and Dr. Sarah L. Symons
- Proposed use of graphene oxide layer as secondary layer to commercially-available zerovalent iron layer used in permeable reactive barriers
- Published abstract in Undergraduate Research in Nature and Clinical Science and Technology (URNCST) conference abstract book (https://doi.org/10.26685/urncst.47) and in the University of Ottawa Science Undergraduate Research Journal (OSURJ) (https://doi.org/10.18192/osurj.v1i1.3705)

Research Assistant *(January 2018 – June 2018)*
McMaster University, Hamilton, ON

- Conducted research in Dr. André Bédard's lab group in the Department of Biology, studying cellular processes regulated by the Lipid Damage Response
- Prepared cycling (sub-confluent) chicken embryo fibroblast (CEF) cells for testing effects of ENPP1 and TG2 enzyme inhibitors on cell proliferation in conditions of normoxia and hypoxia
- Employed techniques of cellular and molecular biology (including tissue culture, proliferation assay, Bradford assay, Western blot)

iSci Research Projects *(September 2017 – April 2020)*
McMaster University, Hamilton, ON

- Light, the Universe, and Everything (Winter 2020)
  Project Leads: Dr. Sarah Symons, Dr. Duncan O'Dell
  - Constructed infographic about Nu Scorpii, a septenary star system in the constellation Scorpius, highlighting interesting facts on stellar multiplicity
  - Designed and presented two seminars on stellar multiplicity to iSci peers using the BOPPPS (Bridge, Objective, Pre-assessment, Participatory learning, Post-assessment, Summary) pedagogical model
- Climate Change (Fall 2019)
  Project Leads: Dr. Jason Brodeur
  - Wrote article and public executive summary on human engineering as a solution to the climate crisis, discussing biology and ethics of induced meat intolerance and reduced human size
  - Recorded a 20-min radio-style podcast presenting human engineering to the public in an engaging and informative manner

- Forensics (Fall 2019)
  Project Leads: Russ Ellis, Dr. Chad Harvey
  - Examined the use of forensic accounting in the downfall of Al Capone
  - Wrote a legal report in defence of Steven Avery, presenting corrupt investigators and scientists as indicators of a frame-up
- Wine Science (Fall 2019)
  Project Leads: Dr. Carolyn Eyles, Russ Ellis
  - Wrote an article reviewing the implications of nanoparticle-conjugated resveratrol particles in wine on enhancing the bioavailability of resveratrol towards a chemotherapeutic end (http://hdl.handle.net/11375/25404)
- Thermodynamics (Winter 2019)
  Project Leads: Dr. Randall Dumont
  - Solved thermodynamics problems involving fire pistons and non-ideal solutions, using mathematical principles, thermodynamic relationships, and Python
- History of the Earth (Winter 2019)
  Project Leads: Dr. Carolyn Eyles, Dr. Sarah Symons
  - Wrote book section on the Cretaceous-Paleogene dinosaur extinction by asteroid impact, presenting both a historical and geological perspective on the Alvarez asteroid impact theory, while considering alternate hypotheses
- Drug Discovery (Fall 2018)
  Project Leads: Dr. Rosa da Silva, Russ Ellis
  - Identified key results from the discovery and development pipeline of the respiratory medicine Ventolin in the form of a literature review
  - Made 5-minute educational YouTube video on memantine, an off-label drug for Obsessive Compulsive Disorder, covering potential mechanisms of action and health effects
- Neuroscience (Fall 2018)
  Project Leads: Dr. Deda Gillespie, Russ Ellis
  - Designed unit on neural development for high school science students, covering structures, differentiation, connections, and disorders
  - Wrote manuals for instructors and students that together provide scientific background, an annotated reading list, a glossary, worksheets, and description of interactive learning activities
- Plant-Animal Interactions (Fall 2018)
  Project Leads: Dr. Chad Harvey
  - Designed and performed an experiment to study the pesticidal role of tannins in the *Myzus persicae* (green peach aphid) x *Arabidopsis thaliana* (thale cress) system; findings suggested that application of tannins does not significantly detriment aphids, but may increase *Arabidopsis* yield in the absence of herbivory

- Cancer – A 21st Century Plague? (Winter 2018)
  Project Leads: Dr. Chad Harvey, Dr. Tom Farrell
    - Reviewed literature on glioblastoma multiforme (brain cancer), discussing current advances pertaining to prevention, detection, and treatment
    - Presented research proposal investigating transmissible non-human cancer, Tasmanian Devil Facial Tumour Disease (DFTD), using NetLogo agent-based modelling software at Synthesis 2018
- Sustainable Energy in Challenging Environments (Winter 2018)
  Project Leads: Dr. Chad Harvey, Russ Ellis
    - Proposed, developed, and extensively tested potential sustainable energy technologies in an integrated laboratory setting
    - Proposed wave farm technology for making sustainable contribution to energy requirements of Town of Petty Harbour-Maddox Cove in Newfoundland and Labrador
- Drugs, Doses, and Biodistribution (Fall 2017)
  Project Leads: Dr. Chad Harvey, Russ Ellis
    - Examined patterns of consumption and absorption of common drugs, including caffeine, alcohol, and tetrahydrocannabinol (THC)
    - Formulated literature review on effects of THC on stress in university students; findings suggested that despite stereotypical association with decrease in stress, THC increases stress
- Planetary Exploration (Fall 2017)
  Project Leads: Dr. Chad Harvey, Russ Ellis
    - Explored physical and orbital properties of single-star planet GJ 273 b in relation to developing Earth-like life using Maple mathematical modelling
    - Designed experiment for studying exoplanet analogue Mars in search of signs of life, proposing Raman spectroscopy for identifying carotenoids
    - Calculated propulsion chemistry and constructed comprehensive plan of three-stage rocket travel to Mars
- Contamination of the Campus Environment? (Fall 2017)
  Project Leads: Dr. Chad Harvey, Dr. Sarah Symons
    - Conducted environmental assessment of McMaster campus area, examining physical, biological, and hydrological features on campus
    - Collected and identified aquatic fauna (bioindicators) within Cootes Paradise as a proxy for environmental health

Co-curricular Experience

Abstract Reviewer *(March 2020)*
Synthesis 2020 iSci Student Symposium, Integrated Science, McMaster University

- Reviewed abstracts written by 2nd-year, 3rd-year, and 4th-year iSci students in submission to Synthesis 2020
- Edited, scored, and gave constructive criticism with reference to content, clarity, tone, grammatical errors, and word limit

Logistics Committee Member *(December 2018 – April 2019)*
Synthesis 2019 iSci Student Symposium, Integrated Science, McMaster University

- Collaborated with members of Logistics Committee to plan fundraising and sponsorship efforts to generate funds for Synthesis 2019
- Sold donuts from Donut Monster in areas and at times of high student traffic on the McMaster University campus
- Outlined sponsorship package and wrote template email for sponsorship requests to be sent to local businesses and organizations

2nd-Year iSci Big Sib *(September 2018 – April 2019)*
Big Sib Program, Integrated Science, McMaster University

- Engaged with 1st-Year iSci Lil Sib through weekly meetings and regular online communication, discussing transition to university and progress in courses
- Advised on iSci research projects and term work, sharing insight regarding instructor and teaching assistant expectations
- Shared extracurricular opportunities relevant to interests of Lil Sib

iSci-For-A-Day Host *(January 2018)*
iSci-For-A-Day Program, Integrated Science, McMaster University

- Hosted Grade 12 student interested in experiencing the Integrated Science (iSci) undergraduate program – student was able to attend Integrated Concept Seminars, an Invited Speaker Seminar, and the Nuclear Reactor Tour that is part of the "Sustainable Energy in Challenging Environments" project
- Discussed iSci program and career interests with student over lunch, making comparisons to other undergraduate programs, and describing career-specific skills gained in iSci
- Toured student around McMaster University buildings, residences, and food places, giving a comprehensive sense of the campus environment

Extra-curricular Experience

WW19 MacSci Sponsorship Coordinator *(February 2019 – December 2019)*
Science Student Representatives (Sciclones), Faculty of Science, McMaster University

- Created sponsorship package that summarizes Welcome Week vision, and attracts sponsorship opportunities by highlighting a menu of marketing outlets
- Contacted and met with past and potential sponsors to outline business relationship (e.g. monetary contribution in exchange for advertising)
- Collaborated with Sciclones Executive Team to help plan Welcome Week in an engaging, inclusive, and accessible manner for ~1700 science students
- Facilitated a welcoming environment for incoming 1st-year students through numerous orientation events during Welcome Week

SciNapse Mentor *(November 2019)*
SciNapse at McMaster, McMaster University

- Mentored McMaster University teams participating in the SciNapse Undergraduate Science Case Competition
- Helped maximize effectiveness of research proposals, sharing ideas to improve convincingness of introductions, and clarity and feasibility of methods
- Answered questions regarding my experience with the competition, with reference to reviewers' comments on my team's research proposal

WW18 MacSci General Representative *(March 2018 – December 2018)*
Science Student Representatives (Sciclones), Faculty of Science, McMaster University

- Facilitated a welcoming environment for incoming 1st-year students through orientation events regarding academics and student life during Welcome Week
- Connected with potential students at open-house events (May@Mac, Fall Preview), answering questions about available programs, extracurriculars, etc.
- Created signs and decorations as a member of the Arts Committee
- Contacted businesses/organizations as a member of the Sponsorship Committee

Lead Alto, Saxophone Section Leader *(September 2017 – April 2018)*
McMaster University Jazz Band, School of the Arts, McMaster University

- Attended weekly rehearsals led by director Mike Malone, established jazz musician and educator, practicing alongside group of 19 musicians
- Prepared jazz repertoire comprising music from eras of swing, bebop, modal, and fusion jazz; as well, prepared set of original material for improvised solos
- Performed at two end-of-term concerts at L.R. Wilson Concert Hall to large audiences of students and parents

Lead Alto, Saxophone Section Leader *(September 2016 – July 2017)*
JAZZ.FM91 Youth Big Band, Toronto, ON

- Performed challenging, modern repertoire alongside internationally-recognized musicians Randy Brecker and Adam James, collaborating in a formal ambience
- Received expert instruction from musician and educator Jules Estrin as well as regular workshops from University of Toronto and Humber College's award-winning faculty Gordon Foote, Pat LaBarbera, and Denny Christianson
- Shaped sound of saxophone section, interpreting and conveying information from conductor to other saxophone players as well as other band members
- Featured in an article by Ori Dagan in the WholeNote magazine

Bandleader *(September 2014 – June 2017)*
The Friday Club (Student-Run Band), Turner Fenton Secondary School

- Collaborated with band members, teacher supervisors, and the school's music council executive team to perform at multiple sold-out annual concerts
- Initiated musical exercises for band members to develop important ensemble skills such as blend, allowing versatility to play different styles and exposing both band members and audience to diverse musical genres
- Delegated tasks to band members to efficiently complete tasks such as composition and arranging punctually and with ease
- Mentored members in performance, improvisation, transcription, composition, and arranging, preparing members for independent musical communication
- Created succession plan to allow comfortable transition for next president & bandleader

Grade 12 Representative *(September 2016 – June 2017)*
Music Council, Turner Fenton Secondary School

- Contributed ideas regarding concerts / musical events during weekly meetings, helping plan event themes, logistics, and to discuss finances
- Utilized social media to market concerts and events organized by Music Council to Grade 12 students, attracting large audiences to consistently sold-out venues
- Effectively multi-tasked during concerts/events, punctually guiding musical groups to stage, directing a small committee of volunteers, and performing

Leadership Developer (Coordinateur) *(February 2015 – March 2015)*
Imaginons Le Français Conférence (IFC), Turner Fenton Secondary School

- Set up activity stations related to France such as Creole bucket-drumming and making Parisian crêpes to emphasize diversity of French peoples
- Worked alongside one co-Leadership Developer to guide group of delegates through French-themed conference to activity stations, highlighting global significance of French language and culture
- Underscored speaking French throughout entire conference, compelling delegates to experiment, learn from mistakes, and acquire new language

Team Member and Regional Competitor *(September 2014 – December 2014)*
Cross-Country Running Team, Turner Fenton Secondary School

- Trained for long-distance running with running specialists and physical education teachers with considerable experience in running sports
- Committed to challenging weekly routine, attending daily after-school training sessions, boosting running speed, stamina, and endurance as well as mental strength to push oneself
- Competed in several 5-7 km races, representing Turner Fenton Secondary School regionally, placing in top 100 out of 1000's of students

Assistant Instructor *(July 2014 – August 2014)*
Aquatics Program, Cassie Campbell Community Centre, Brampton, ON

- Assisted aquatics program instructors with teaching swim strokes and life-saving techniques to children, adults, and seniors
- Trained and received certification for SFA (Standard First Aid) with CPR-C (Cardiopulmonary Resuscitation) and AED (Automated External Defibrillator)
- Completed additional training sessions to gain pool-specific interpersonal skills (varying for children and adults) in learn-to-swim programs

Team Member and Regional Competitor *(October 2013 – February 2014)*
Swim Team, Turner Fenton Secondary School

- Trained under international-level triathlon expert Laura Gazzola, developing individual- and team-specific swim skills
- Competed in Region of Peel Secondary School Athletic Association (ROPSSAA) Swim Meet, representing Turner Fenton Secondary School regionally, scoring bronze medal in team relay
- Gained independent and group competitive experience through several swim meets, taking place at swim clubs such as Kiwanis Cobra Swim Club and institutions such as McMaster University, broadening network with other athletes of different ages and growing as a competitor

Presenter *(October 2013 – October 2013)*
Peel Children's Water Festival, Brampton, ON

- Delivered multiple 10-15 min presentations to school children, teachers, and parents on importance of healthy water for the Canada Goose
- Obtained resources/props to aid presentation, adding significant pictorial value to information delivered through speech, clarifying concepts for audience

Skills

Communication

- Collection, organization, and formal presentation of information in professional research media (reports, articles, presentations, posters, podcasts, videos)
- Languages (English, Gujarati, Hindi)

Computer

- Microsoft suite (Word, Excel, PowerPoint, Publisher, OneNote, Teams)
- Data analysis and visualization (Prism, Excel, Python)
- Bioinformatics (Geneious, Benchling)
- Mathematical modelling (Maple)
- Agent-based modelling (NetLogo)
- Molecular docking (PyMol)
- Image analysis (ImageJ)

Microbiology, Molecular Biology, Plant Biology

- Media preparation
- Molecular cloning
- Genomic DNA (gDNA) extraction
- Plasmid DNA extraction (miniprep)
- Primer design
- Polymerase chain reaction
- Restriction digestion
- Agarose gel electrophoresis
- Conjugation
- Transduction (ΦM12)
- Electroporation (*E. coli* EPI500)
- Site-specific recombination (Flp-*FRT*)
- Yeast recombineering (LiAc/SS/PEG)
- Seed scarification and sterilization
- Nodule kinetics assay
- Shoot dry weight assay
- Acetylene reduction assay
- Competition for nodule occupancy assay

Laboratory Instruments

- Light microscope
- pH metre
- Centrifuge
- Incubator
- -80 ^{o}C freezer
- -20 ^{o}C freezer
- UV-Vis spectrophotometer
- Mass balance
- Autoclave
- Micropipette
- Serological pipette
- Biosafety cabinet (level 1, level 2)
- Fume hood
- Electroporator

<u>Regulatory Training</u>

- Accessibility for Ontarians with Disabilities Act (AODA) and Human Rights Code
- Asbestos Awareness
- Autoclave Awareness Training
- Biosafety Training
- COVID-19 Awareness
- Chemical Handling and Spills
- Ergonomics
- Faculty of Health Sciences (FHS) Hospital Fire Safety
- Fire Safety
- Gas Cylinder
- Health and Safety Orientation
- Primer on Privacy
- Slips, Trips, and Falls
- Violence and Harassment Prevention
- WHMIS 2015

<u>Certifications</u>

- Ontario Secondary School Diploma (OSSD)
- International Baccalaureate (IB) Diploma
- McMaster University School of Interdisciplinary Science Teaching Assistant Development Program Completion
- safeTALK: Suicide Alertness For Everyone

- SFA (Standard First Aid) + CPR-C (Cardiopulmonary Resuscitation) + AED (Automated External Defibrillator)
- Lifesaving Society Programs: Bronze Star, Bronze Medallion (Waterfront Supervisory Certification), Bronze Cross (Assistant Lifeguard Certificate)
- Canadian Red Cross (Lifesaving Society): Learn-to-Swim Levels 1-9 (All)
- Canadian Red Cross: Babysitting Course
- G Driver's Licence

Online Courses

- Biology Meets Programming: Bioinformatics for Beginners
  by Pavel Pevzner and Phillip Compeau
  Coursera, May 2020

Conferences (attended, unless specified otherwise)

- Designing Canada – The Masterclass, by The Socrates Project, 2020
  - McMaster University, Hamilton, ON
- Synthesis: Integrated Science Student Symposium, 2019
  - McMaster University, Hamilton, ON
- MedX Hamilton, 2019
  - McMaster University, Hamilton, ON
- Synthesis: Integrated Science Student Symposium, 2018
  - Presented ISCI 2A18 "Cancer – A 21st Century Plague?" project on investigating transmissible non-human cancer, Tasmanian Devil Facial Tumour Disease (DFTD), using NetLogo agent-based modelling
  - McMaster University, Hamilton, ON
- IntRO: Introduction to Research Opportunities, 2018
  - McMaster University, Hamilton, ON
- Imaginons Le Français Conférence (IFC), 2015
  - Led group of delegates as Coordinateur
  - Turner Fenton Secondary School, Brampton, ON
- START: Students, The Active Risk-Takers, 2014
  - Turner Fenton Secondary School, Brampton, ON
- IBMC: International Baccalaureate Mentorship Conference, 2013
  - Turner Fenton Secondary School, Brampton, ON

Societies

- Canadian Society of Environmental Biologists
  - Member since October 2020
- Canadian Society of Petroleum Geologists
  - Member since February 2019

Interests

- Rhizobium-legume symbiosis
    - Symbiotic nitrogen fixation, nodulation, bacteroid differentiation
    - Engineering nitrogen fixation in non-legumes
    - Evolution of plant-microbe symbioses
- Music: engagement with music on multiple levels, including performance, composition, and leading bands; skilled at several instruments, including alto saxophone, flute, clarinet, and piano

Personal Projects

- Directory of rhizobium scientists
- Catalogue of nodulation genes
- YouTube playlists of talks/seminars on rhizobium, SynBio, systems biology
- Website featuring music education resources, mainly music theory